

THE STRUCTURE OF MATHEMATICAL REALITY

Phase I — Chapter 3

The Map of Mathematical Fields

Note Template

Self-Study Curriculum

How to use this template

Complete these notes **after** the reading and the lecture. The goal is not to copy phrases from the text — it is to reconstruct the ideas in your own language. Leave every item blank on first pass; fill it in from memory, then check.

1. For each field, state the central question in your own words

Algebra:

Analysis:

Geometry:

Topology:

Number theory:

Combinatorics:

Logic:

Probability:

2. Three bridges between fields that I now understand

a. _____ + _____ = _____

Why this bridge is powerful:

b. _____ + _____ = _____

Why this bridge is powerful:

c. _____ + _____ = _____

Why this bridge is powerful:

3. The hierarchy of structure preservation means

More structure $\rightarrow$ more/fewer situations it applies to $\rightarrow$ more/fewer things you can prove

Less structure $\rightarrow$ _____ $\rightarrow$ _____

4. An example from my own work where I used a concept from one

field but could have asked a question about it from another field:

describe it

5. The part of the map I now want to explore further

which border or field, and why
